Dynamic Chemicals

Dynamic by Nature





Products/Services

Water Treatment Chemicals

Boiler Water Treatment
RO Water Treatment
Cooling Tower Treatment
RO Membrane Treatment
ETP Polyelectrolytes

Fireside Chemical and Fuel Additives

Industrial Water Treatment

De alkalisation (De Scaling Process)
Softener Process
Demineralisation

IBR Boiler Manufacturing

Water Wall Membrane

ID Fan And FD Fan

MS Chimney

Industrial RO (Reverse Osmosis)

Water Softener





In Partnership with J.S. Fabricators

(Govt. Approved Boiler Manufacturer)

Boiler & Its Types

IBR BOILER:

IBR BOILER IS CLOSED VESSEL THAT IS USED TO PRODUCE STEAM THERMAL ENERGY RELEASED BY COMBUSTION OF FUEL IS UTILIZED TO MAKE STEAM AT DESIRED PRESSURE AND TEMPERATURE

भारतीय बायलर विनियम (IBR) के मुताबिक, IBR Boiler का मतलब है 25 लीटर से ज्यादा क्षमता वाला कोई भी बंद बर्तन जिसका इस्तेमाल ताप (Temp), दवाव (Pressure) में भाप (Steam) बनाने के लिए किया जाता है।

ADVANTAGE OF IBR BOILER

NO RISK OF EXPLOSION: MANUFACTURING FROM IBR MATERIAL

SAFETY ASSURANCE: GOVERNMENT CERTIFIED INSPECITON



There are mainly two types of boiler

1. Water Tube Boiler
2. Fire Tube Boiler (Smoke Tube)
3. Smoke Cum Water Tube

Water Tube Boiler : एक प्रकार का बॉयलर है जिसमें पानी आग से बाहरी रूप से गर्म की गई नलियों में घूमता है।

Fire Tube Boiler : फायर -ट्यूब बॉयलर एक प्रकार का बॉयलर है जिसमें गर्म गैसे आग से एक या एक से अधिक ट्यूबों के माध्यम से पानी के एक सीलबंद कंटेनर से होकर गुजरती है। गैसों की गर्मी तापीय चालन द्वारा ट्यूबों की दीवारों के माध्यम से स्थानांतरित होती

Main Components of Combustion boiler:-
Burner, Combustion chamber, heat exchanger, Supply lines and Return line, Circulatory pumps



Boiler Scales

Scales are formed out of salts and Impurities present in the water

Scales are maily made up of calcium carbonate (CaCO), Calcium sulphate (CaSO4), Barium Sulphate (BaSO4), Calcium Fluoride (CaF), Calcium Phosphate [Ca3(PO4)

Disadvantages of Boiler Scales

1. Wastage of Fuel: The rate of heat transfer from boiler to inside water is greatly reduced. This causes increase in fuel consumption. इंधन को बर्बादी

2. Corrosion जंग

3. Boiler Becomes Unsafe बॉयलर असुरक्षित हो जाता है।

4. Decrease in Efficiency of Boiler बॉयलर में दक्षता की कमी

5. Danger of Explosion विस्फोट का खतरा

BOILER SCALE INSULATING PROPERTIES

Scale Thickness in inches	Average Efficiency Loss	Oil Wasted Per 1000 Gallons	Coal Wasted per Ton	Gas Wasted per 1000 Cubic Feet
1/64	4%	40	80#	40
1/32	7%	70	140#	70
1/16	11%	110	220#	110
1/8	18%	180	360#	180
3/16	27%	270	540#	270
1/4	38%	380	760#	380
3/8	48%	480	960#	480
1/2	60%	600	1200#	600
5/8	75%	740	1480#	740
3/4	90%	900	1800#	900

Removal of Scales & Sludges

Scales can be removed with the help of Antiscalant Chemical, Blow down Operations, EDTA Solution.



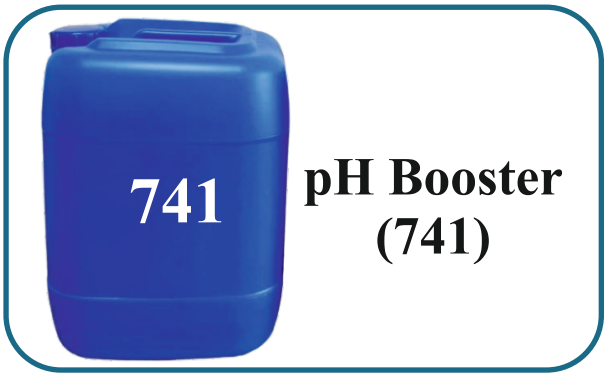
Boiler feed water treatment chemicals

1. Antiscalant Chemicals: It is used to remove scale and sludge mainly.

Antiscalant compound is a dianionic polyelectrolyte (DAPE), or more particularly poly [disodium 3-(N,N-diallylamino)propanephosphonate]



2. Oxygen Scavenger: It is used to remove Oxygen and reduce the danger of boiler tube failure or excessive feed water leaks.



3. pH Booster: It counteracts the fall in pH of water due to removal of salts and reduces Corrosion.



4. Fireside Powder: Used to Increase Temperature in furnace भट्ठी में तापमान बढ़ाना

2. Reduce the formation of slag धातुमल

3. Reduce the formation of CARBON

RO(Reverse Osmosis)

Reverse Osmosis (RO):- Used to remove dissolved minerals & reduce TDS and make it suitable for used in feed water tanks in industries.



ROAntiscalants :- Highly effective in preventing the membranes from scaling.

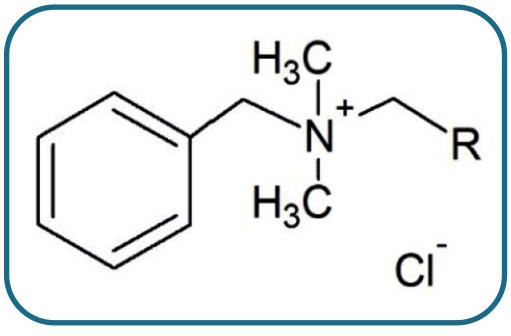
pH Booster: It counteracts the fall in pH in the permeate water due to removal of salts.

Cooling Tower



Cooling Tower:-A Cooling tower is a specialized heat exchange in which air and water are brught into direct contact with each other.

Cooling Tower Antiscalant:- Dynamic chemical's Atiscalant can be used as disinfectant to display microorganisms and prevent recurring growth.



Biocides:- Biocides are used to control bacteria in cooling tower system.

Anticorrosion Chemicals:- Used to prevent Corrosion

Descaling Process

Descaling: Descaling is the process of removing scale from equipment such as boilers and heat exchangers. It involves the removal of thick layers of oxides formed on metals at elevated temperatures. Descaling safely and effectively removes Limescale; a hard, off-white, chalky deposit that can impair the operation of various components.

SCALE DENSITY **Vs** ACTION PLAN

Scale Density Ranges and Required Action

Scale Density g/ft2 or (mg/cm2)	Recommended Action
< 23 (25)	No action required
23 – 70 (25 – 75)	Chemically clean within one year
70 – 93 (75 – 100)	Chemically clean within three months
> 93 (100)	Chemically clean before further operation

PHILOSOPHY OF TREATMENT/PROCESS OF DE-SCALING.

Dynamic chemicals has been manufacturing safe chemical solutions for years, so we are familiar with all shapes, sizes and boiler configurations. The procedure outline below is brief and simple for most boiler cleanings.

1. Shut down boiler.
2. Blow down with pressure still on.
3. Drain and flush unit.
4. Close steam valve.
5. Add DYNAMIC CHEMICALS SOLUTION, and then top off with water.
6. Circulate through the drain or low point and return from the pressure relief valve or high point of the boiler to a vented container.
7. Circulate for 14-18 hours depending on volume.
8. Open steam valve.
9. Drain, flush, re-fill and start up.

DESCALING CHEMICALS

Conditioning Chemical
D-Anti Scalent Chemical
Neutralizer Chemical



New Plant and Machinery Setup



Semi/Fully Automatic 5 KLD to 500 KLD Small, Medium and Large scale effluent treatment plants. (STP)



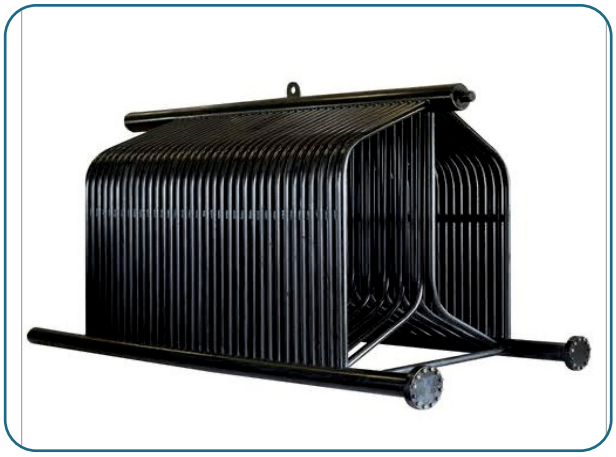
New RO Setup from 100 LPH to maximum size.
- High quality membrane
- Activated Graular Carbon
- High Quality Filter & Media



New Water Softner Installation.



New Boilers of Every Type



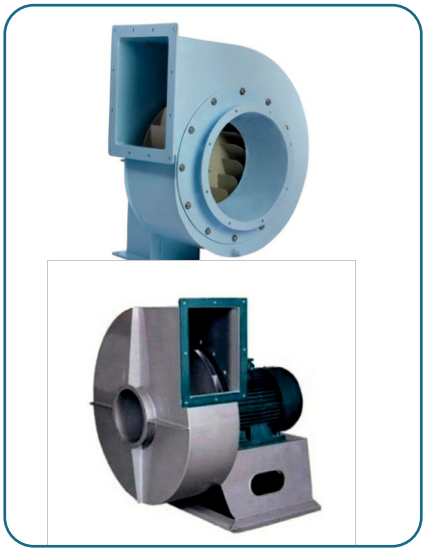
Water Wall Membrane



Boiler Feed Pump



Activaed Carbon in RO Plants



ID FAN And FD Fan



MS Chimney
(As per Required)

Dynamic Chemicals

Dynamic by Nature

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